SRNet: Semantic Anchoring and Fine-Grained Refinement Network for Semantic Change Detection in Remote Sensing Images

Zhenghao Jiang, Xiaoqian Zhu, Wanxia Deng, Lei Liu, DongShuo Yin, *Member, IEEE*, Biao Wang, and Li Liu, *Senior Member, IEEE*

***Abstract*—Semantic change detection (SCD) identifies changed regions and semantic transitions in bitemporal remote sensing images, supporting applications such as urban expansion monitoring and land use analysis. Prior studies have achieved notable progress under full parameter training. However, inadequate global semantic consistency and blurred local feature refinement persist. This *semantic drift* remains a fundamental obstacle to advancing SCD. In this work, we propose the semantic anchoring and fine-grained refinement network (SRNet), a novel parameter-efficient fine-tuning (PEFT) framework. SRNet mitigates *semantic drift* through a two-stage process of semantic anchoring and fine-grained refinement. In particular, we introduce a multiscale fine-grained adapter (MA) to adapt pretrained features to the SCD task, achieving semantic anchoring that suppresses cross-temporal feature drift. A fine-grained spatio-temporal module (FGSM) is then designed to capture subtle local change information across temporal images. We evaluated SRNet on several representative datasets that span various land-cover scenarios, including urban (SECOND, Hi-UCD-mini), rural (HRSCD), and wetland (SJH-SCD) environments. On HRSCD and Hi-UCD-mini, it attains 64.80% and 56.14% F_{scd}, outperforming the current state-of-the-art models by 3.67% and 6.19%, respectively. Cross-domain transfer from urban to wetland scenes further validates its generalization ability. The code will be available at https://github.com/XiaoJ058/RS-SCD**

***Index Terms*—Fine-grained, parameter-efficient fine-tuning (PEFT), remote sensing images, semantic anchoring, semantic change detection (SCD).**

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Zhenghao Jiang is with the College of Meteorology and Oceanography, National University of Defense Technology, Changsha 410073, China (e-mail: jiangzhenghao26@nudt.edu.cn).

Xiaoqian Zhu is with the Army Engineering University, Nanjing 210007, China (e-mail: zhu_xiaoqian@sina.com).

Wanxia Deng and Lei Liu are with the College of Meteorology and Oceanography and the Key Laboratory of High Impact Weather (special) of CMA, National University of Defense Technology, Changsha 410073, China (e-mail: dengwanxia14@nudt.edu.cn; liulei17c@nudt.edu.cn).

DongShuo Yin is with the School of Artificial Intelligence and Data Science, University of Science and Technology of China, Hefei 230026, China (e-mail: yinds@mail.tsinghua.edu.cn).

Biao Wang is with the School of Resources and Environmental Engineering, Anhui University, Hefei 230601, China (e-mail: wangbiao-rs@ahu.edu.cn).

Li Liu is with the College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: liuli_nudt@nudt.edu.cn).

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I. Introduction

REMOTE sensing change detection (CD) aims to identify land surface changes from multitemporal satellite imagery [1], [2], [3], [11], [12], [13], serving as a fundamental tool for spatio-temporal Earth observation. Depending on the level of semantic interpretation, CD can be broadly categorized into binary CD (BCD) and semantic CD (SCD) [4], [5], [6], [7], [8], [14]. BCD focuses solely on discriminating changed pixels from unchanged ones, answering *whether* a change has occurred but offering no information on *what* the transition entails [15], [20], [21], [61], [62]. This semantic blindness limits its practical value in applications that demand interpretable land-cover dynamics, such as differentiating urban encroachment from seasonal vegetation cycles [19], [22], [23].

By contrast, SCD jointly localizes changed regions and identifies their land-cover categories both before and after the change [25], [26], [27], [28]. It explicitly models the transition of semantic classes, yielding a class-aware characterization of surface evolution. This capability makes SCD particularly suitable for semantically sensitive scenarios, including postdisaster damage grading, urban renewal monitoring, and precision agriculture assessment.

Early SCD methods primarily adopted the postclassification comparison strategy [24], often combined with traditional machine learning classifiers such as support vector machines or random forests [16], [17], [18]. These approaches typically model change information by constructing multichannel feature inputs. While they achieved moderate success on low-resolution imagery, their capacity to capture deep semantic correlations across both spatial and temporal dimensions remains limited when applied to very-high-resolution scenes. With the advancement of deep learning, end-to-end trainable architectures have become the dominant paradigm in remote sensing image analysis. Their advantages are threefold: strong feature extraction, flexible task modeling, and natural adaptability to multimodal data. This trend has seen rapid development in object detection [53], semantic fusion [51], and mask-based CD [9], [10]. As illustrated in Fig. 1, representative backbones include convolutional neural networks (CNNs),

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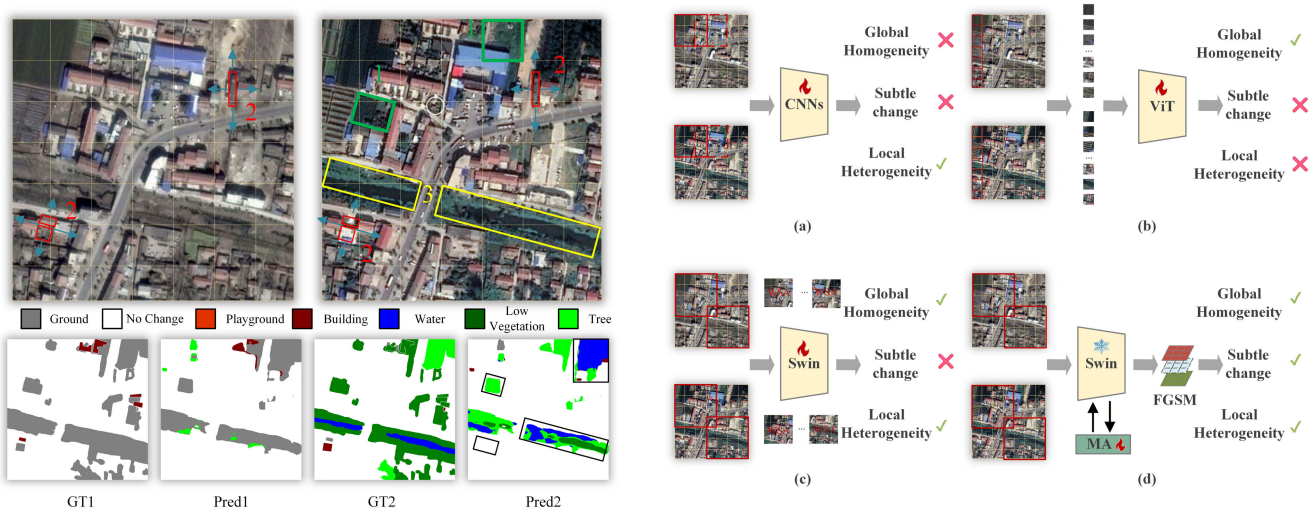


Fig. 1. Illustration of four representative semantic change detection frameworks: (a) CNN-based paradigm; (b) ViT-based paradigm; (c) Swin-ViT-based paradigm; and (d) the Swin-based fine-tuning paradigm. Three typical challenges are also presented: 1) globally abundant homogeneous information; 2) subtle-change regions that are susceptible to environmental interference; and 3) locally heterogeneous regions with high semantic similarity

vision Transformers (ViTs), and more recently, Mamba-based models [7], [29], [32], [33], [45], [49].

CNNs were among the first deep learning architectures applied to the SCD task. Benefiting from their inherent translational equivariance and local receptive fields, CNNs excel at capturing fine-grained textural patterns and spatial details, which are essential for delineating precise change boundaries in high-resolution imagery. Early efforts along this direction include HRSCD [23], which formulated SCD as a multiclass semantic segmentation problem by feeding concatenated bitemporal images into a CNN backbone. While this end-to-end paradigm achieved substantial improvements over traditional postclassification methods, its simple input stacking strategy offered limited explicit modeling of cross-temporal object correspondences. To better capture the evolutionary relationships of ground objects across time, subsequent works introduced siamese-based architectures. Representative models such as SCDNet and SSCDL [1], [25] employ dual-branch encoders with shared weights or contrastive learning objectives to extract temporally consistent features, thereby enhancing the discrimination of changed regions. Furthermore, to strengthen the multiscale representational capacity of CNN encoders, methods like DFINet and BiSRNet [25], [27] incorporate attention-like mechanisms or feature pyramid designs within the convolutional framework. Despite these advances, the intrinsic locality of the convolution operation fundamentally constrains the receptive field, limiting the ability of CNN-based SCD models to perceive large-scale contextual changes and long-range semantic dependencies across the entire scene.

In contrast to CNNs, ViTs offer an alternative modeling paradigm by leveraging global self-attention to capture long-range spatial dependencies [32], [52], [53], [54]. This global receptive field enables ViTs to effectively perceive large-scale contextual changes that are often missed by convolution-based encoders. Recognizing this advantage, several recent SCD methods have sought to incorporate Transformer-style architectures into their frameworks. For instance, BGSNet [34] adopts a ViT-based backbone to enhance the global semantic modeling of bitemporal features. SCNet [30] employs a dual-branch design where the SAM2 backbone (built upon the multiscale ViT architecture) operates in parallel with a CNN branch, aiming to fuse local details with global context. SCanNet [25] adopts a serial strategy, cascading a CNN encoder with a CSwin Transformer to progressively refine the feature representations. While these hybrid designs partially alleviate the limited receptive field inherent to pure CNNs, they introduce two notable complications. First, the structural series connection or parallelization of heterogeneous backbones substantially increases computational overhead, making large-scale training and inference less efficient. Second, and more critically, the coupling of local and global branches raises a challenging optimization tradeoff. In the low-data regime typical of SCD, the model favors the rapidly converging CNN branch, which readily captures dominant local textures. Consequently, the ViT branch may receive insufficient gradient updates, leading to underutilized global contextual information and suboptimal cross-temporal change modeling [44]. These issues underscore the need for an architecture that reconciles local precision with global awareness. Swin Transformer [39] meets this requirement via hierarchical window-based self-attention, offering a strong foundation for bitemporal feature extraction in SCD. However, full parameter training of the Swin model on downstream SCD tasks substantially disturbs the visual priors learned during pretraining. This degradation directly weakens the model's ability to extract stable features across T1 and T2 [36], [38], [46]. This degradation directly undermines the model's ability to extract deep invariant features.

More recently, with the rapid development of large-scale pretrained vision models, parameter-efficient fine-tuning (PEFT) has emerged as a promising paradigm for adapting powerful visual representations to downstream tasks with only a small number of trainable parameters [37], [47]. Instead

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of fully updating the backbone, PEFT seeks to preserve the general visual priors learned from large-scale pretraining while introducing lightweight task-specific modules for efficient adaptation. This property is particularly attractive for SCD, where annotated bitemporal samples are usually limited, and full fine-tuning can easily lead to overfitting or degradation of pretrained representations.

However, the direct application of existing PEFT strategies to SCD remains nontrivial. Unlike conventional image classification or single-temporal semantic segmentation, SCD jointly involves semantic understanding and change discrimination. It requires not only accurate local detail perception and global contextual reasoning but also strong cross-temporal feature association between bitemporal images. Existing adapter-based PEFT methods are mainly designed for single-image recognition or segmentation tasks, and often lack explicit mechanisms for modeling spatial structure and temporal correspondence [55], [56], [57], [58], [59], [60]. As a result, when applied to complex remote sensing scenes, the pretrained feature space may be disturbed by the coexistence of global semantic homogeneity and local structural heterogeneity, leading to *semantic drift*. Such drift weakens the semantic consistency of pretrained representations and further degrades the model's ability to distinguish true semantic changes from irrelevant appearance variations.

To address these limitations and promote the effective use of PEFT in SCD, this work proposes a PEFT framework, termed semantic anchoring and fine-grained refinement network (SRNet). The key motivation of SRNet is to preserve the semantic priors of the pretrained backbone while enhancing its adaptability to fine-grained cross-temporal change modeling. In particular, SRNet mitigates semantic drift in two steps. It first anchors the high-level semantics inherited from the pretrained backbone, and then refines fine-grained spatial details for more accurate change discrimination. The main contributions of this work are summarized as follows.

1) *A Multiscale Fine-Grained Adapter (MA):* We propose MA for lightweight feature adaptation with a frozen pretrained backbone. It treats pretrained features as semantic anchors and adapts downstream representations through adaptive input calibration, multiscale local detail modeling, and output semantic consistency matching. In this way, it preserves stable semantic priors while improving the adaptation to complex remote sensing scenes, thus alleviating semantic drift during fine-tuning.
2) *A Fine-Grained Spatio-temporal Module (FGSM):* We propose FGSM for precise cross-temporal change refinement. After the pretrained semantics are anchored, FGSM further strengthens the modeling of subtle local changes through a spatial-difference branch and a spatio-temporal multiscale branch. The former explicitly enhances local feature discrepancies between bitemporal images, while the latter captures change patterns across different spatial scales. This design enables the model to better distinguish fine-grained semantic changes from irrelevant appearance variations, thereby improving the accuracy of SCD in complex remote sensing scenes.
3) Extensive experiments across four SCD datasets covering urban (SECOND, Hi-UCD-mini), rural (HRSCD), and wetland (SJH-SCD) scenes demonstrate SRNet's advantage over existing methods. Cross-domain generalization from SECOND to urban scenes and to the SJH-SCD-7 dataset further validates the robustness and transferability of the proposed parameter-efficient adaptation strategy.

The remainder of this article is organized as follows. Section II details the SRNet architecture. Section III describes the experimental setup and datasets. Section IV presents results and analysis. Section V concludes this article.

II. Methodology

In this section, we first present the formal formulation of the SCD problem, followed by a detailed introduction of the proposed SRNet. The proposed network specifically focuses on preserving global homogeneous consistency while enhancing the discrimination of local heterogeneous features. Accordingly, we will elaborate on the design principles of the MA and the FGSM, respectively. The overall framework of SRNet is illustrated in Fig. 2, and the processing pipeline comprises the following two core stages.

1) *Domain Knowledge Injection Stage:* By freezing the Swin Transformer backbone to preserve the general feature extraction capabilities acquired during pretraining, we design the MA to inject task-specific domain knowledge tailored for the SCD scenario.
2) *Local Change Discrepancy Refinement Stage:* The spatial-difference branch explicitly captures surface variation cues, while the spatio-temporal aggregation branch jointly leverages spatial and temporal contextual information to further discriminate fine-grained local changes.

These two stages are executed sequentially and function in a complementary manner, collectively enabling the network to effectively address the semantic drift issue and generate high-precision SCD results.

A. SCD Problem Formulation

We address the semantic drift problem in SCD. Due to variations in illumination, seasonality, and human activities, identical land-cover classes often exhibit spectral discrepancies across time, while distinct classes may appear similar at local boundaries. This phenomenon weakens the model's ability to discriminate pseudochanges and identify true change types, requiring a capability to jointly perceive global semantics and local details. Consequently, the SCD task aims to: given a bitemporal image pair of the same area, the model must simultaneously localize changed regions and accurately recognize the specific change categories.

Formally, we denote the bitemporal input images as $(T_1, T_2 \in \mathbb{R}^{H\times W\times 3})$. For feature extraction, SRNet employs Swin-Base as the backbone, whereas SRNet-tiny adopts Swin-Tiny. In both variants, the BatchNorm layers and the MA adapters in the Swin encoder are unfrozen and optimized during training, while all remaining parameters of the Swin

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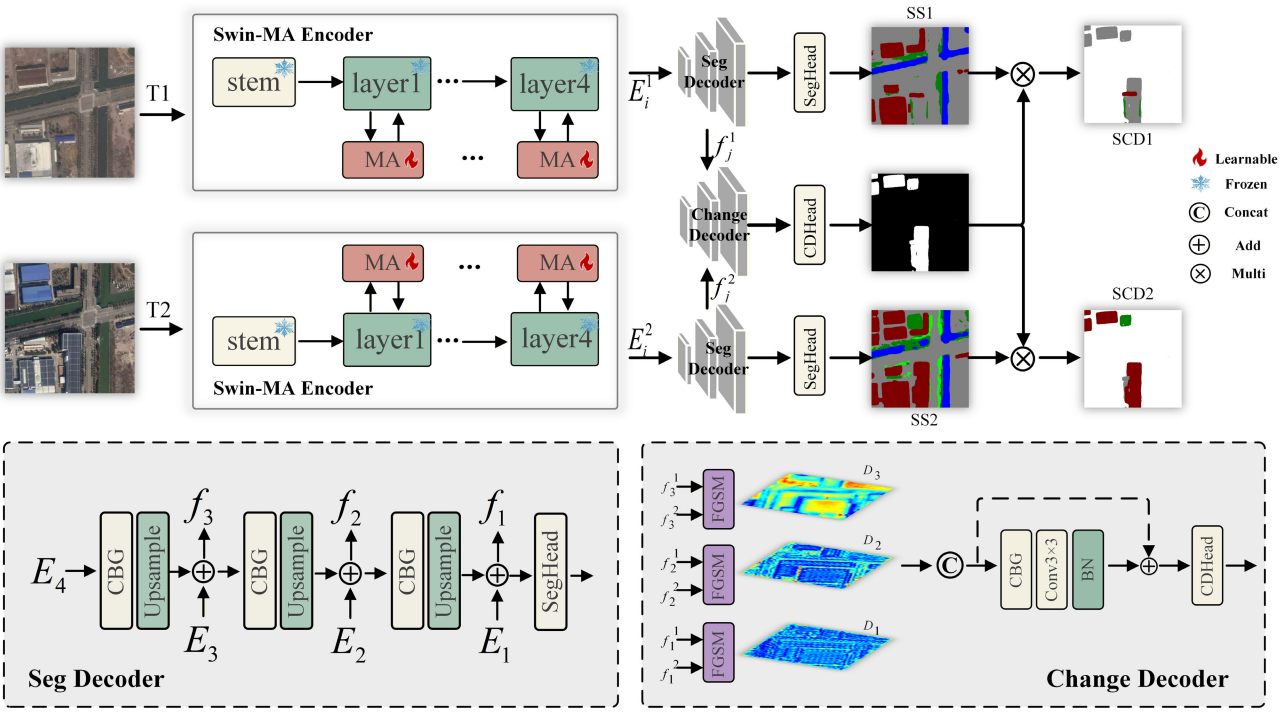


Fig. 2. Framework of the proposed SRNet. CBG denotes a composite block of 3 × 3 convolution, batch normalization, and GELU activation.

backbone are frozen. This parameter-efficiently fine-tuned backbone is referred to as the Swin-MA Encoder, whose multiscale feature maps at each layer are denoted as E_i^1, E_i^2 ($i = 1, 2, 3, 4$). In the decoding stage, according to task decomposition, the decoder responsible for land-cover semantic recognition is defined as the Seg Decoder, whose output multiscale features f_j^1, f_j^2 ($j = 1, 2, 3$) are subsequently utilized as follows: the feature pairs f_j^1, f_j^2 are fed into the FGSM for change information extraction, while the highest resolution features $f_1^1, f_1^2 \in$ and $\mathbb{R}^{H/4\times W/4\times 64}$ are input into the SegHead to generate the bitemporal semantic segmentation maps, denoted as SS1 and SS2, respectively. In the change decoder stage, the feature pairs f_j^1, f_j^2 are further processed within the FGSM to enhance cross-temporal discrepancy cues. Inside the module, a spatial-difference branch $\mathcal{S}_j$ and a multiscale spatio-temporal aggregation branch T_j operate in parallel to capture fine-grained local change information; their outputs are then fused along the channel dimension to yield the enhanced change feature $\mathbf{D}_j$. Subsequently, the multiscale features are progressively integrated through a series of lightweight residual blocks and passed into the CDHead to produce the final binary change map, denoted as BCD. Finally, by performing pixelwise multiplication of BCD with SS_1 and SS_2, we obtain the complete SCD results for the two temporal phases, which are defined as SCD_1 and SCD_2, respectively.

B. Multiscale Fine-Grained Adapter Module

Existing fine-tuning methods for image semantic information extraction predominantly rely on fully connected adapter structures inherited from NLP [48]. However, these adapters do not explicitly model 2-D spatial topology, scale-varying remote sensing objects, or bitemporal feature interaction, making them less suitable for SCD. Some approaches attempt to address this limitation by freezing the foundation model and introducing fully trained convolutional branches for shallow feature matching. Although effective to a certain degree, such auxiliary-branch designs introduce extra architectural complexity and require fusion between heterogeneous foundation models and CNN features. To address this issue without destabilizing the backbone's global semantic anchoring capability, the proposed MA is designed as a spatially aware lightweight adapter that sequentially follows the frozen window self-attention (WSA). As illustrated in Fig. 3, the Swin-MA Encoder employs a Swin-base architecture stacked with {2, 2, 18, 2} Swin-MA Blocks across its four layers. Significantly, by keeping the WSA weights frozen, the encoder retains its ability to recognize stable object-level semantics.

Let $\mathbf{x} \in \mathbb{R}^{B\times L\times C}$ be the input to an MA module, where B is the batch size, $L = H \times W$ is the spatial size, and C is the channel dimension. The MA module operates as follows:

$$\mathbf{x}_{\text{fuse}} = S_1 \odot \text{LayerNorm}(\mathbf{x}) + S_2 \odot \mathbf{x} \tag{1}$$

$$\mathbf{z} = \text{Linear}_{C\to 64}(\mathbf{x}_{\text{fuse}}), \quad \mathbf{z}_{2D} = \text{Reshape}(\mathbf{z}) \in \mathbb{R}^{B\times 64\times H\times W} \tag{2}$$

where $\odot$ denotes channelwise multiplication. Two learnable vectors $S_1, S_2 \in \mathbb{R}^C$ balance the normalized and original features: when the domain shift is small, S_2 dominates and preserves the pretrained semantic anchors; when the shift is large, S_1 increases and normalization compensates. Equation (2) projects the stabilized features into a compact 64-D bottleneck

$$\mathbf{r} = \frac{1}{3}\sum_{k\in\{1,3,5\}} \text{DWConv}_{k\times k}(\mathbf{z}_{2D}) + \mathbf{z}_{2D} \tag{3}$$

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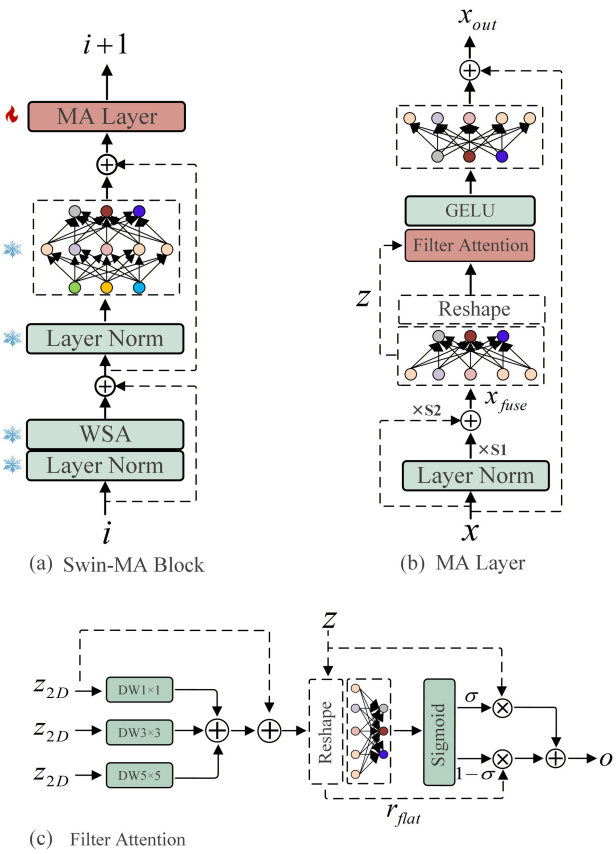


Fig. 3. Overview of the proposed Swin-MA block. (a) Overall architecture of the Swin-MA block, where WSA denotes the window-based self-attention module in the Swin Transformer. (b) Detailed structure of the MA module. (c) Detailed structure of the filter attention module.

$$\mathbf{g} = \sigma\left(\mathrm{MLP}_{128\to 64\to 64}\left([\mathbf{z}, \mathbf{r}_{\mathrm{flat}}]\right)\right). \tag{4}$$

Equation (3) performs multiscale intratemporal refinement through three parallel depthwise convolutions with a residual connection, ensuring that the adapter refines rather than replaces backbone features. Equation (4) defines the task-aware gate: the concatenation $[\mathbf{z}, \mathbf{r}_{\mathrm{flat}}]$ is passed through a two-layer MLP with sigmoid activation to produce a channel-wise gating vector $\mathbf{g} \in [0,1]^{64}$, which interpolates between the original projection and the multiscale residual. $[\cdot,\cdot]$ denotes channelwise concatenation, σ is the sigmoid function, and $\mathbf{r}_{\mathrm{flat}} \in \mathbb{R}^{B\times L\times 64}$ is the refined feature reshaped back from $\mathbf{r}$

$$\mathbf{o} = \mathbf{g} \odot \mathbf{z} + (1-\mathbf{g}) \odot \mathbf{r}_{\mathrm{flat}} \tag{5}$$

$$\mathbf{x}_{\mathrm{out}} = \mathbf{x} + \mathrm{Dropout}\left(\mathrm{Linear}_{64\to C}\left(\mathrm{GELU}\left(\mathbf{o}\right)\right)\right). \tag{6}$$

Equations (5) and (6) define the gated fusion and output projection. Equation (5) combines the original projection and the multiscale residual via the gate $\mathbf{g}$. Equation (6) projects the gated features back to dimension C via a residual connection. After MA refinement, the T1 and T2 features are semantically aligned and distributionally consistent. These drift-suppressed representations are then fed into FGSM, where intertemporal comparison models genuine land-cover transitions on a stable feature foundation.

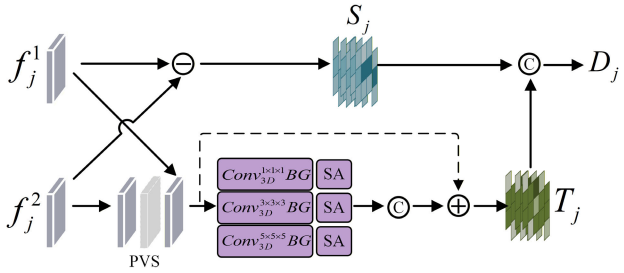


Fig. 4. Architecture of the FGSM, where PVS denotes pseudovideo sequence. BG denotes batch normalization, and GELU activation.

C. Fine-Grained Spatio-Temporal Module

While the Swin-MA Encoder establishes robust global semantic anchors to suppress pseudochanges induced by intraclass spectral variance, SCD further demands the precise localization of fine-grained temporal transitions at the pixel level. A straightforward strategy adopted by several existing methods is to construct pseudovideos directly from raw images and apply 3-D convolutions to model temporal evolution. However, raw images contain abundant low-level appearance variations, such as illumination changes, shadows, and seasonal effects. Organizing these appearance fluctuations into a continuous transition can cause regions with only superficial changes to be misinterpreted as meaningful semantic transitions, thereby increasing false alarms.

By contrast, FGSM performs temporal modeling on the semantically stabilized features produced by MA, where the pretrained backbone's category-level understanding is preserved, and appearance-induced perturbations are substantially reduced. As illustrated in Fig. 4, FGSM further departs from such single-branch designs through a dual-branch architecture. In addition to a spatio-temporal branch that uses 3-D convolutions to capture interframe feature evolution, FGSM incorporates an explicit spatial discrepancy branch that directly computes per-scale feature differences, providing complementary detection of gradual transitions and instantaneous changes. A residual refinement module fuses the outputs of both branches across three scales, allowing the network to selectively enhance change-relevant features while suppressing noise. This multiscale temporal modeling, performed at three decoder resolutions, captures fine-grained and coarse structural changes simultaneously, whereas the aforementioned single-branch 3-D approaches lack dedicated spatial comparison and multiscale temporal aggregation. In essence, FGSM explicitly models the collaborative evolution of spatial structure and temporal semantics. The formulation proceeds as follows:

$$S_j = \left| f_1^j - f_2^j \right| \tag{7}$$

$$T_j = C_{3\mathrm{D}}^{(k)}\left(\mathcal{V}\left(f_1^j,\ f_2^j\right)\right), \quad j \in \{1,2,3\}, \quad k \in \{1,3,5\} \tag{8}$$

$$T_j = \mathrm{Cat}\left(\mathrm{Mean}\left(T_j\right),\ \mathrm{Max}\left(T_j\right)\right) \tag{9}$$

$$f_{\mathrm{BCD}} = \mathrm{Res}\left(\mathrm{Cat}\left(S_j, T_j\right)\right). \tag{10}$$

Central to this module is the temporal modeling operator $\mathcal{V}$, which explicitly constructs a pseudovideo tensor of shape $\mathbb{R}^{B\times C\times T\times H\times W}$ by stacking the bitemporal features

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along a newly inserted temporal axis. This structural reformatting enables the subsequent multiscale 3-D depthwise separable convolutions $C_{3\text{-D}}^{(k)}$ (with kernel sizes $k \in 1, 3, 5$) to jointly apprehend what has changed and how it has evolved. Unlike simple frame differencing [see (7)], which only captures magnitude shifts, the 3-D convolutions extract motion-aware semantic cues from the local spatio-temporal neighborhood.

To consolidate these fine-grained temporal patterns into a globally coherent change signature, a spatial aggregation (SA) mechanism is employed. By concatenating the mean and max pooled responses along the temporal channel, the module distills both the dominant trend and the most salient deviation within the pseudovideo sequence. Finally, the multiscale difference features S_j and the aggregated temporal features T_j are fused at the channel level and passed through a residual refinement block. This hierarchical fusion effectively compensates for the edge blurriness inherent in coarse semantic anchors, sharpening the final binary change delineation and enhancing the localization accuracy of small or fragmented changed regions.

D. Network Training

We employ the binary cross-entropy (BCE) loss to optimize the BCD output and the standard cross-entropy (CE) loss to supervise semantic segmentation. Since SCD annotations provide a single label per pixel encoding both change status and semantic category, we derive the binary change label from the SCD annotation for BCE loss computation. For the CE loss, only pixels labeled as changed are included; unchanged pixels are masked out during gradient back-propagation. Under this formulation, changed pixels receive explicit supervision to predict their correct prechange and postchange semantic categories, which naturally drives their feature representations apart. Conversely, unchanged pixels receive no CE gradient signal, allowing the shared backbone and frozen pretrained weights to maintain feature consistency across time phases without an explicit consistency loss. The overall training objective is formulated as follows:

$$\mathcal{L}_{bcd} = -y_c \log(p_c) - (1 - y_c) \log(1 - p_c) \tag{11}$$

$$\mathcal{L}_{ss} = -\frac{1}{N} \sum_{i=1}^{N} y_i \log(p_i) \tag{12}$$

where y_c denotes the ground-truth binary label indicating changed regions and p_c is the predicted change probability at each pixel. N represents the total number of land-cover and land use (LCLU) categories in the dataset. For the ith semantic class, y_i is the ground-truth label and p_i is the predicted probability.

III. Experiment Details

A. Metrics

We assess SCD performance using three complementary metrics: mean intersection over union (mIoU), the separation Kappa (SeK) coefficient, and F_{scd} [25]. As a standard pixel-level agreement measure in CD, mIoU quantifies the overlap between predicted maps and ground-truth annotations

$$IoU_c = \frac{TP}{TP + FP + FN} \tag{13}$$

$$IoU_{uc} = \frac{TN}{TN + FP + FN} \tag{14}$$

$$mIoU = \frac{IoU_c + IoU_{uc}}{2}. \tag{15}$$

where subscripts c and uc index changed and unchanged pixels, respectively. In the associated confusion matrix, true positives (TPs) and false negatives (FNs) record correctly and incorrectly classified changed samples, while true negatives (TNs) and false positives (FPs) track unchanged counterparts. The SeK metric, an adaptation of Cohen's Kappa, offers additional diagnostic value: it diminishes as the confusion matrix deviates from symmetry, thereby penalizing classifiers with skewed prediction tendencies

$$kappa = \frac{p_0 - p_e}{1 - p_e} \tag{16}$$

$$SeK = e^{IoU_c - 1} \times kappa. \tag{17}$$

In this expression, p_0 is the observed accuracy and p_e denotes the expected agreement computed from row and column marginals.

Let $Q \in \mathbb{R}^{(N+1)\times(N+1)}$ denote the confusion matrix over N semantic change categories plus the unchanged class (indexed as class 0). Element q_{ij} records the number of pixels whose ground-truth label is class j and whose predicted label is class i, where $i, j \in \{0, 1, \ldots, N\}$. The F_{scd} metric isolates classification quality within changed areas by excluding the unchanged class from both the precision numerator and the recall reference. Its components are defined as

$$P_{\text{scd}} = \frac{\sum_{i=1}^{N} q_{ii}}{\sum_{i=1}^{N} \sum_{j=0}^{N} q_{ij}}, \quad R_{\text{scd}} = \frac{\sum_{i=1}^{N} q_{ii}}{\sum_{i=0}^{N} \sum_{j=1}^{N} q_{ij}} \tag{18}$$

$$F_{\text{scd}} = \frac{2 \cdot P_{\text{scd}} \cdot R_{\text{scd}}}{P_{\text{scd}} + R_{\text{scd}}}. \tag{19}$$

The numerator $\sum_{i=1}^{N} q_{ii}$ counts correctly classified changed pixels (predicted class i matches ground-truth class i, $i \neq 0$), while the denominator $\sum_{i=1}^{N} \sum_{j=0}^{N} q_{ij}$ sums over all pixels *predicted* as a changed class ($\forall i \neq 0$), regardless of their true label. Thus P_{scd} measures the precision of semantic classification *within predicted change regions*. While the denominator $\sum_{i=0}^{N} \sum_{j=1}^{N} q_{ij}$ sums over all pixels *truly labeled* as changed ($\forall j \neq 0$). Thus, R_{scd} measures the recall of semantic classification *within ground-truth change regions*. The harmonic mean F_{scd} merges both quantities to assess the accuracy of land-cover transition segmentation within changed areas, complementary to the region-level mIoU and the edge-sensitive SeK.

B. Datasets

Fig. 5 provides a concise overview of the four SCD datasets, covering urban (SECOND, Hi-UCD-mini), rural (HRSCD), and wetland (SJH-SCD) scenarios. This diverse combination

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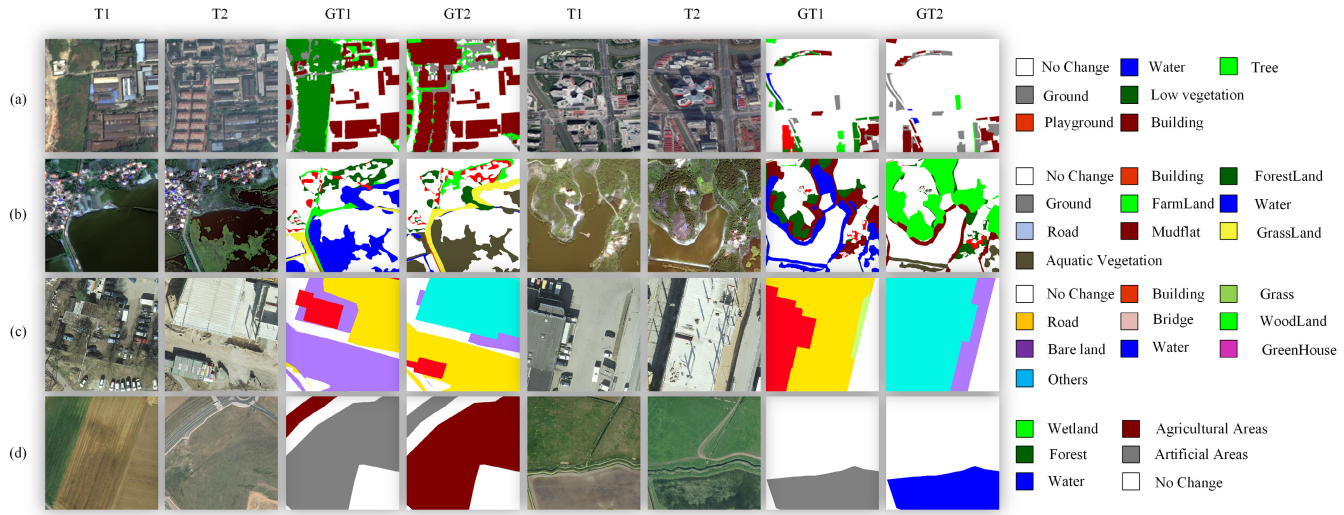


Fig. 5. (a)–(d) SECOND, SJH-SCD, Hi-UCD-mini, and HRSCD datasets, respectively. T1, T2, GT1, and GT2, respectively, represent the pretemporal image, posttemporal image, and the ground truth of change.

of typical change types allows for a comprehensive evaluation of model performance and generalization across different data scales and application contexts.

The SECOND dataset [43] comprises six major land-cover classes—water, tree, ground, low vegetation, playground, and building—each playing a significant role in urban construction. It contains 4662 image pairs with resolutions ranging from 0.5 to 3 m and a uniform size of 512×512 pixels. Following the official split, the dataset is partitioned into training, validation, and test sets of 2670, 296, and 1696 image pairs, respectively.

The SJH-SCD dataset is a self-constructed dataset for wetland SCD. It consists of 1468 bitemporal image pairs acquired by the Gaofen-1/2/6 satellites over Shengjin Lake, China. The images, with resolutions of 1 and 2 m, are cropped into 512×512 nonoverlapping patches. The dataset covers land-cover transitions including water, bare ground, roads, farmland, mudflats, grassland, forestland, buildings, and aquatic vegetation. It is divided into 1174 training pairs and 294 testing pairs. We plan to release it publicly in the near future with standardized splits, evaluation protocols, and class-definition documentation.

The Hi-UCD mini dataset [42] includes ten land-cover classes—water, grass, building, greenhouse, road, bridge, bare land, woodland, others, and an unlabeled category—at a spatial resolution of 0.1 m. To improve computational efficiency, we adopt a preprocessing strategy similar to EGMSNet [28], cropping images into 512×512 patches and discarding unchanged pairs. This yields final datasets of 571, 100, and 705 image pairs for training, validation, and testing, respectively.

The HRSCD dataset [23] is derived from the BD ORTHO database, France's national high-resolution aerial orthoimagery repository. It contains 291 coregistered RGB image pairs acquired in 2006 and 2012, focusing on five land cover change categories: artificial areas, agricultural areas, forest, wetlands, and water. The images were cropped into 512×512 patches and partitioned into training, validation, and test sets comprising 7573, 946, and 946 pairs, respectively.

C. Implementation Details

All experiments were conducted on a workstation equipped with an Intel Core i5-12600KF CPU and an NVIDIA GeForce RTX 4070 Ti Super GPU (16 GB memory). The proposed model was implemented in PyTorch [31]. We trained the model for 50 epochs with a batch size of 4, an initial learning rate of 3×10^{-4}, and a weight decay of 3×10^{-4}, using the AdamW optimizer with a cosine annealing learning rate schedule [41]. To improve generalization, on-the-fly data augmentation consisting of random flipping and rotation was applied during training. For fair comparison, all competing models were evaluated under the same input resolution, data augmentation pipeline, and number of training epochs.

IV. Experiment Analysis

A. Overall Performance

As shown in Table I, the compared methods span four architectural paradigms. CNN-based methods (HRSCD4, SSCDl, BiSRNet, DFINet, TBFFNet, BTSCD) employ ResNet34 or custom CNN encoder–decoders and are fully fine-tuned from pretrained ImageNet [40] weights. Hybrid CNN-Transformer methods include SCanNet, which augments a ResNet34 backbone with a MixTransformer for cross-temporal feature interaction, and SCNet, which integrates a SAM2 Hiera-based ViT encoder alongside a ResNet34 branch. Mamba-based methods include ChangeMamba, for which we adopt the small variant in our experiments. Transformer-based PEFT methods (SRNet and SRNet-tiny) freeze a Swin Transformer backbone and insert lightweight MA modules, updating only 4.80–5.54-M parameters while preserving pretrained semantic structure. SRNet achieves competitive performance on the SECOND dataset with only 5.54-M trainable parameters, reaching 74.41% mIoU, 25.43% SeK, and 65.08% F_{scd}. Fig. 6 reports the per-class performance of the competing methods. SRNet achieves the highest scores in five out of

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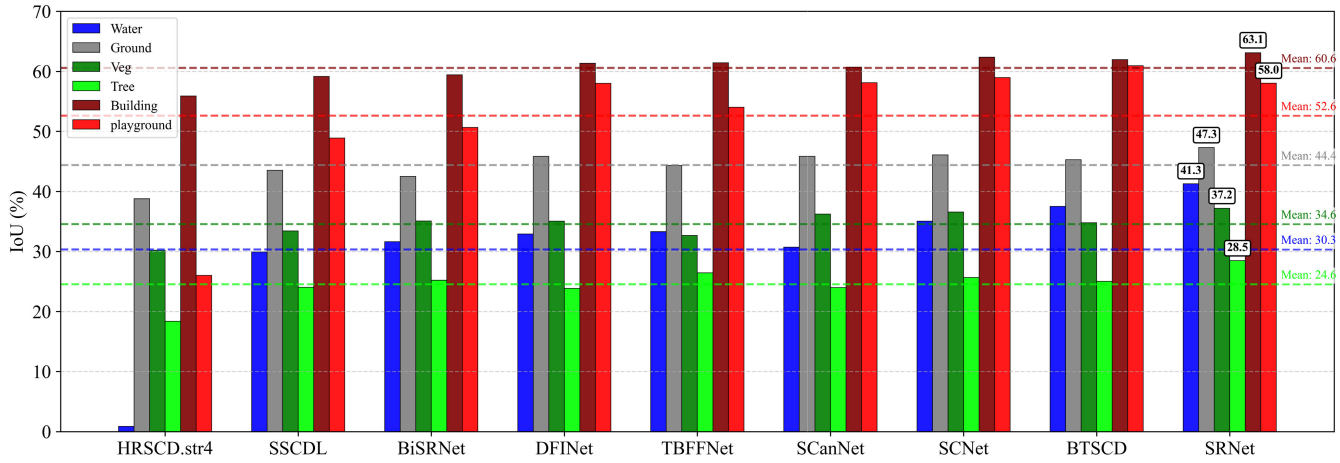


Fig. 6. Comparison of different models on each class of the SECOND dataset.

TABLE I
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON FOUR SCD DATASETS

Method	SECOND			HRSCD			SJH-SCD			Hi-UCD-mini		
	mIoU (%)	SeK (%)	F_{scd} (%)	mIoU (%)	SeK (%)	F_{scd} (%)	mIoU (%)	SeK (%)	F_{scd} (%)	mIoU (%)	SeK (%)	F_{scd} (%)
HRSCD4 [23]	69.83	17.00	57.01	57.37	5.15	47.25	73.52	23.32	61.44	67.03	9.57	44.48
SSCDL [25]	72.11	21.17	61.61	67.79	10.00	59.63	74.54	26.76	64.94	66.49	10.47	46.08
BiSRNet [25]	72.29	21.41	61.41	68.91	11.07	59.92	74.82	27.38	65.62	64.77	8.11	43.49
DFINet [29]	73.20	23.04	63.39	68.77	11.27	61.13	76.27	31.51	69.41	66.73	9.61	43.16
TBFFNet [33]	73.04	22.42	62.10	67.98	8.35	57.48	74.96	28.65	66.66	67.21	10.85	45.64
BTSCD [35]	73.68	23.73	63.66	63.40	6.58	50.57	75.19	29.55	67.92	67.21	11.53	47.67
ChangeMamba-s [7]	72.73	22.25	62.42	67.53	10.65	57.59	77.87	34.71	71.40	64.17	10.32	46.68
SCanNet [27]	72.77	22.89	63.30	68.22	11.58	60.37	76.14	31.62	69.65	66.89	10.29	45.15
SCNet [30]	74.12	24.44	64.25	68.81	11.18	60.52	77.71	34.49	71.62	69.66	13.95	49.95
SRNet-tiny	73.99	24.26	63.85	70.08	14.77	63.85	77.76	34.21	70.96	69.64	15.49	51.99
SRNet	74.41	25.43	65.08	70.59	15.19	64.80	78.81	36.39	72.36	71.50	18.79	56.14

Note: Red indicates the best; blue indicates the second-best.

TABLE II
COMPARISON OF MODEL COMPLEXITY, TRAINING COST, AND INFERENCE SPEED

Model	Total Params (M)	Trainable Params (M)	FLOPs (G)	Training Time (h)	Inference Speed (Images/s)
BTSCD [35]	24.91	24.91	228.25	6.30	3.22
SCNet [30]	168.46	23.65	807.23	5.25	2.19
SCanNet [27]	27.90	27.90	264.95	5.59	3.31
DFINet [29]	24.26	24.26	208.96	5.61	3.12
ChangeMamba-s [7]	66.16	66.16	164.47	8.07	2.61
BiSRNet [25]	23.38	23.38	190.30	4.96	4.79
SSCDL [25]	23.31	23.31	189.76	4.18	7.81
TBFFNet [33]	26.53	26.53	401.09	6.08	2.94
HRSCD4 [23]	13.71	13.71	43.97	5.40	3.67
SRNet-tiny	33.13	4.80	105.82	4.73	5.67
SRNet	93.62	5.54	229.61	6.82	3.38

the six land-cover categories, demonstrating its effectiveness in detecting changes across both fine-grained classes and large-scale classes. The only exception is the playground class, where it lags marginally behind BTSCD and SCNet, likely due to the relatively limited samples in this category. These results underscore that SRNet, despite being fine-tuned with only a small number of parameters, maintains robust detection capability across diverse object scales and sample sizes. Table II shows the model complexity. SRNet has more total parameters because it uses Swin-Base as the frozen backbone, which itself contains approximately 88 M parameters. On the contrary, SRNet-tiny achieves the best overall balance between inference speed and performance while maintaining a lightweight design; most of its evaluation

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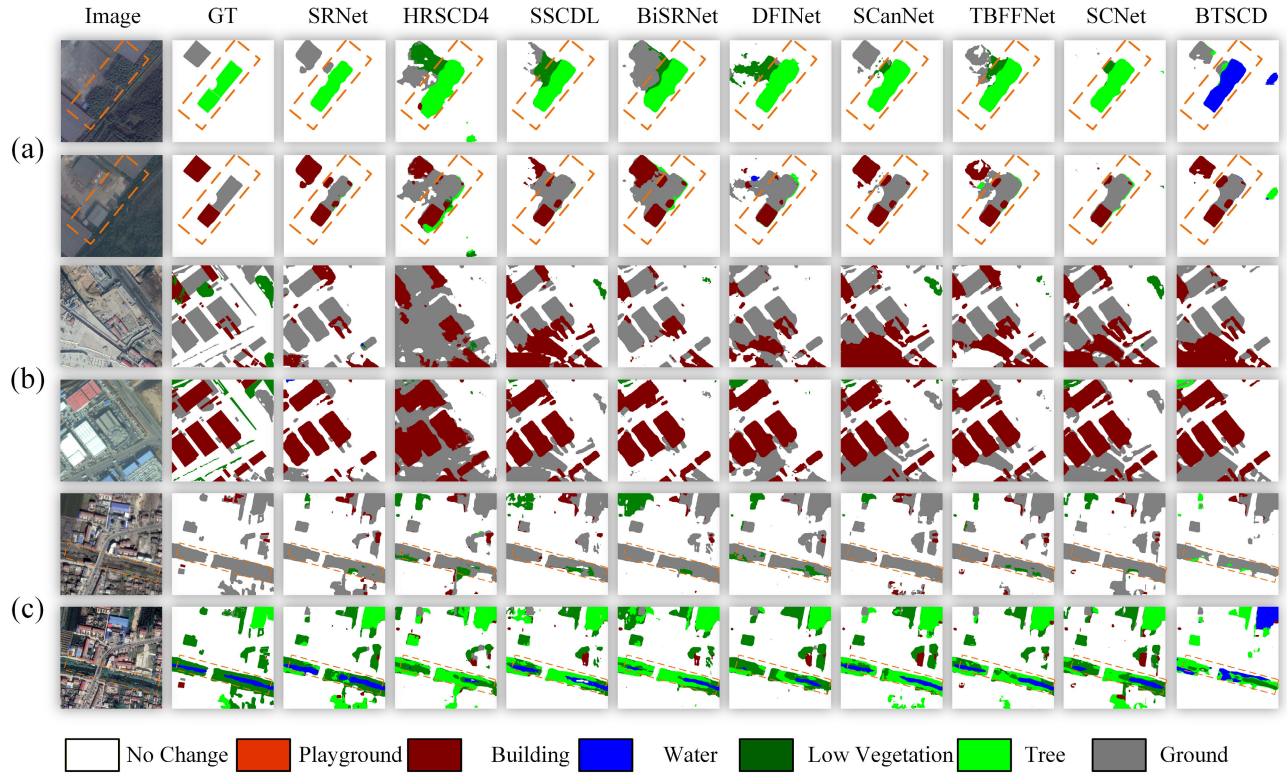


Fig. 7. (a)–(c) Comparative experiment results for different methods on the SECOND dataset, with major differences highlighted in orange rectangles.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT FINE-TUNING METHODS ON THE SECOND DATASET

Model	Input size	Fscd (%)	mIoU (%)	SeK (%)	FLOPs (G)
SRNet	512	65.08	74.41	25.43	229.61
$SAM_{Adapter}$	1024	64.76	73.89	24.60	1285.84
SAM_{LoRA}	512	58.96	70.88	18.93	209.81

metrics are only slightly lower than those of SRNet. The main advantage of PEFT in our framework is not to reduce total parameters or FLOPs, but to maximize model performance and adaptation efficiency by updating only a small number of trainable parameters while keeping most pretrained parameters frozen.

The advantage of SRNet is more evident in data-scarce scenarios: on the Hi-UCD-mini dataset, it outperforms SCNet by 4.84% in SeK, verifying its strong fine-tuning ability and robustness with limited training samples. On the high-resolution, sample-rich HRSCD dataset, SRNet maintains competitive performance and surpasses all baselines across all evaluation metrics. These results confirm that the model adapts well to both low-data fine-tuning and data-rich scenarios. We further evaluate several representative fine-tuning strategies (Table III), including Swin + MA, SAM + Adapter, and SAM + LoRA. Swin + MA achieves the optimal tradeoff between accuracy and computational cost. SAM + Adapter, constrained by its fixed 1024×1024 input resolution, incurs extra upsampling overhead and higher FLOPs during inference; despite improved accuracy, it still underperforms SRNet. On the contrary, SAM + LoRA adapts to varying input sizes by dynamically adjusting positional encodings, reducing computation with only a marginal accuracy loss.

B. General Scenario Analysis

We selected a set of representative scenes encompassing land-cover combinations that occur with high frequency in remote sensing imagery, ranging from pronounced changes to semantically ambiguous transitions. These general scenarios are deliberately chosen to evaluate the detection performance of competing models, as they epitomize the two most prevalent error modes in SCD: false alarms in unchanged regions and misclassification among spectrally similar land-cover types. Figs. 7 and 8 illustrate the performance of SRNet across multiple datasets. The yellow-boxed region in Fig. 7(a) presents a deceptively challenging case. While the transition from bare ground to buildings constitutes a pronounced change, the adjacent unchanged bare ground areas are characterized by complex local textures and irregular boundaries, which are prone to triggering false detections. As observed, the competing models, despite correctly identifying the building emergence, all exhibit varying degrees of false alarms in these static bare ground regions. On the contrary, SRNet effectively suppresses such boundary-induced FPs. This robustness stems from the model's capacity to preserve local detail extraction on par with traditional CNN-based architectures, even as it adopts Swin as the backbone for capturing global context. By maintaining this balance, SRNet remains sensitive to genuine semantic transitions while

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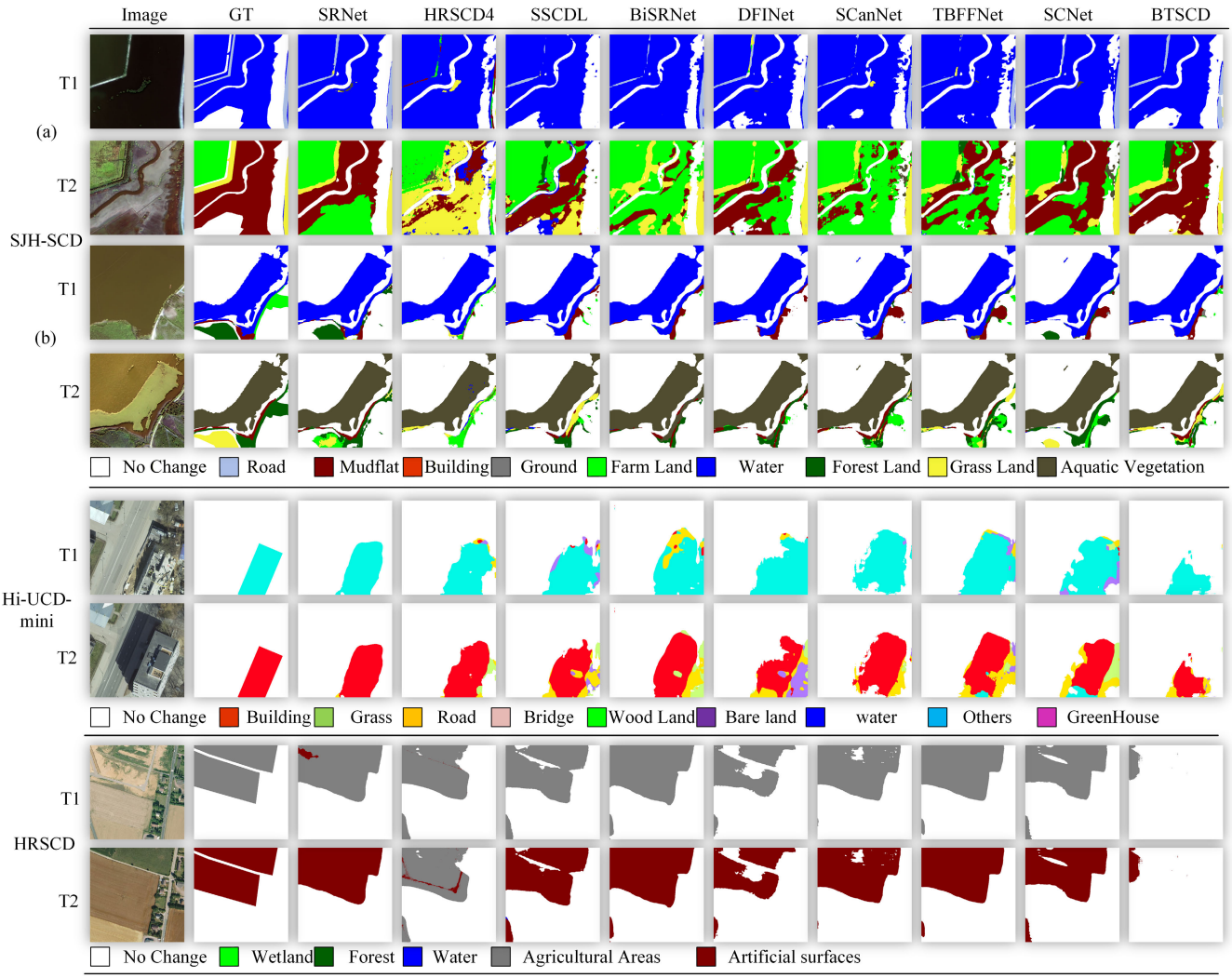


Fig. 8. Comparative experiment results for different methods on the SJH-SCD, Hi-UCD-mini, and HRSCD datasets, where (a) and (b) show the results on the SJH-SCD dataset.

filtering out irrelevant high-frequency spatial noise. Similarly, the yellow-boxed area in Fig. 7(c) features a more challenging composition comprising three easily confusable classes—trees, low vegetation, and water. The subtle spectral and textural gradients among these categories create a high risk of semantic ambiguity, particularly between low vegetation and trees. Indeed, the comparative results show that other methods consistently misclassify significant portions of low vegetation as trees, causing blurred semantic boundaries across the scene. SRNet avoids this confusion, accurately delineating both the low vegetation and water extents. Its stronger feature discrimination stems from an ability to learn fine-grained semantic representations, which helps resolve the interclass ambiguity that hampers conventional approaches in such transitional zones.

On the SJH-SCD dataset, characterized by diverse land-cover categories, subtle interclass differences, and limited training samples, SCD presents a considerable challenge. Under such conditions, existing mainstream SCD methods all exhibit FPs and missed detections to varying extents. On the contrary, SRNet achieves superior overall performance in detecting changes such as those between elongated roads and grassland [as shown in the top-left corner of Fig. 8(a)], as well as between easily confused categories like farmland and grassland [bottom-left corner of Fig. 8(b)]. This advantage can be attributed to its multiscale fine-tuning architecture, which enables effective learning of change features from complex land-cover environments even with scarce training data.

C. Challenging Scenario Analysis

Fig. 9 presents the extraction results of SRNet across four datasets featuring distinct and challenging remote sensing scenarios. In Fig. 9(a), the scene is characterized by ambiguous boundaries between water bodies and adjacent vegetation, compounded by high spectral similarity between these two classes. Under such conditions, conventional CD models often struggle to maintain sharp semantic delineation and are prone to boundary blurring or category confusion. SRNet addresses this challenge through its FGSM in the decoding stage, which explicitly couples cross-temporal change cues and suppresses semantic drift in the feature space. This mechanism refines the decision boundaries between spectrally similar classes,

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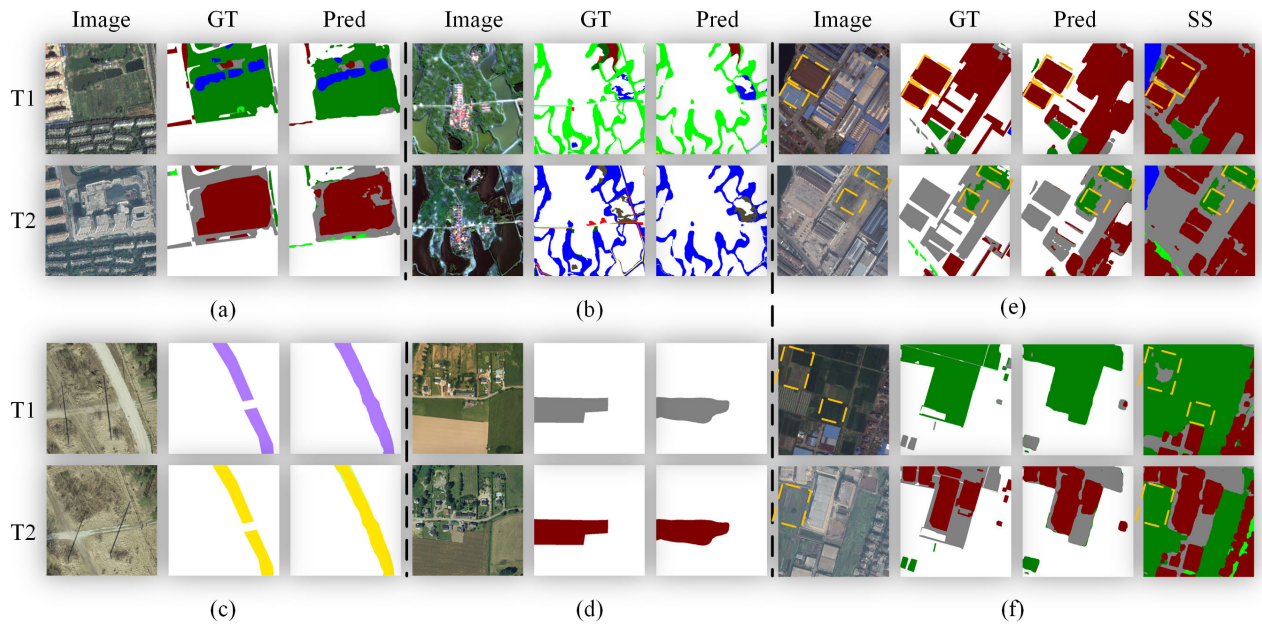


Fig. 9. Visual results of SRNet on (a) SECOND, (b) SJ-SCD, (c) Hi-UCD-mini, and (d) HRSCD datasets, where (e) and (f) show the effectiveness of SRNet in extracting global homogeneous consistency on the SECOND dataset.

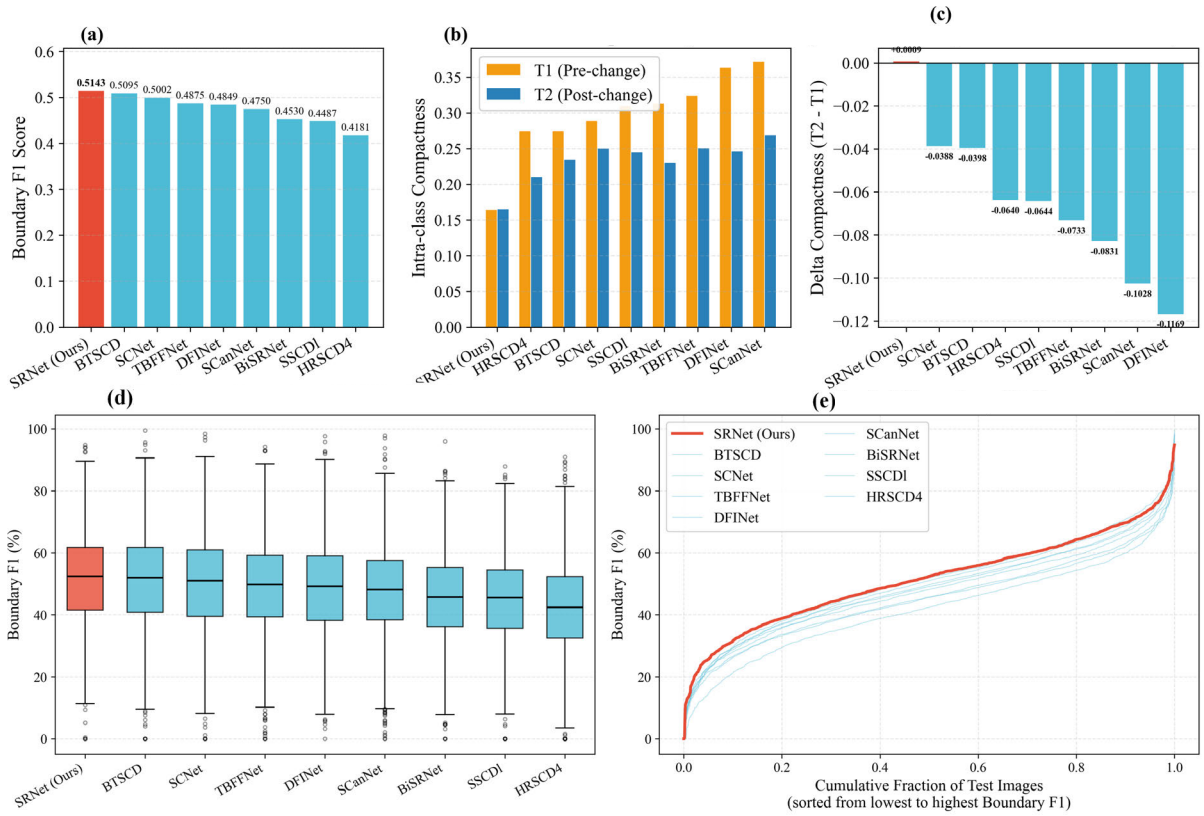


Fig. 10. Comparison of boundary $F1$, intraclass compactness (T1, T2), and temporal feature variation $|\Delta|$ for all evaluated methods on the SECOND dataset. (a) Boundary $F1$ (higher is better). (b) Intraclass compactness (lower is better). (c) T1 $\rightarrow$ T2 feature drift (|Delta| closer to 0 = more robust). (d) Per-image boundary $F1$ distribution. (e) Sorted boundary $F1$ curve.

enabling SRNet to accurately delineate both the spatial extent of changed regions and their precise semantic categories. For the subtle building changes visible in the surrounding area, the model must preserve sharp change edges and maintain the structural integrity of internal land-cover regions. This requirement poses a difficulty for frozen Transformer backbones, which often lack the local inductive bias necessary to capture fine-grained spatial details. SRNet compensates

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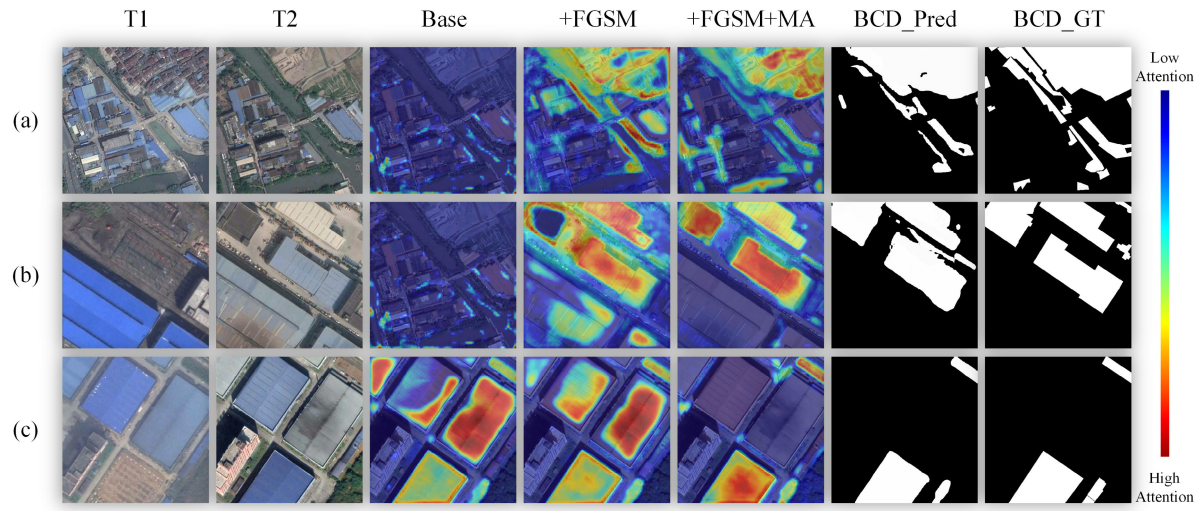


Fig. 11. (a)–(c) Ablation study results of SRNet on the SECOND dataset, where "Base" denotes the baseline model with a fully trained Swin backbone without any adapters. In these visualizations, warmer colors indicate regions where the model exhibits stronger focus.

TABLE IV
QUANTITATIVE COMPARISON OF BOUNDARY $F1$, INTRACLASS COMPACTNESS, AND TEMPORAL FEATURE STABILITY

Model	Boundary F1 ↑	Intra T1 ↓	Intra T2 ↓	$\lvert\Delta\rvert$ (T1→T2) ↓
SRNet (Ours)	**0.5143**	**0.1645**	**0.1653**	**0.0008**
BTSCD	0.5095	0.2749	0.2351	0.0398
SCNet	0.5002	0.2890	0.2502	0.0388
TBFFNet	0.4875	0.3243	0.2510	0.0733
DFINet	0.4849	0.3635	0.2466	0.1169
SCanNet	0.4750	0.3719	0.2691	0.1028
BiSRNet	0.4530	0.3134	0.2302	0.0832
SSCDl	0.4487	0.3095	0.2451	0.0644
HRSCD4	0.4181	0.2748	0.2107	0.0641

for this limitation through the MA, which injects multiscale spatial sensitivity into the frozen Swin hierarchy without disrupting the global semantic priors. By anchoring representations against high-frequency perturbation, the adapter enables SRNet to retain both sharp boundary details and low-frequency textural cues. As a result, the model accurately captures subtle building alterations while suppressing the feature distortion that frequently leads to missed detections or false alarms in unchanged regions. Fig. 9(e) and (f) further illustrates the model's robustness to variations in object appearance across different imaging conditions. As highlighted by the yellow annotations, the visual characteristics of buildings and vegetation differ noticeably across the scenes—a variation that commonly triggers spurious change predictions in less discriminative models. SRNet's consistent semantic separation between these categories can be attributed to the synergy between the adapter and the spatio-temporal module, which together reinforce the stability of the learned representations.

D. Semantic Drift Analysis

To further examine whether SRNet mitigates semantic drift, we provide a dedicated analysis from two aspects: the boundary quality of changed regions and the structure of the semantic feature space. For boundary localization, we first extract boundary pixels from both predicted change masks and ground-truth masks using morphological gradient operators. A tolerance of four pixels is adopted to account for minor annotation and alignment errors. We then compute boundary precision, recall, and $F1$ score for each test image, and report both global statistics and per-image distributions. As shown in Fig. 10(a) and Table IV, SRNet obtains the highest global boundary $F1$ score of 0.5143 among all compared methods. It slightly outperforms BTSCD (0.5095) and shows a clear improvement over HRSCD4 (0.4181). The per-image box plot also shows a higher median and a narrower interquartile range. These results suggest that SRNet can better preserve the semantic consistency of changed regions, especially near object boundaries where semantic drift is more likely to occur.

At the same time, the key to suppressing semantic drift lies in learning feature representations that are both intraclass compact and cross-temporally stable. We extract per-pixel feature vectors from the preclassification layer of each model and apply L2 normalization. Intraclass compactness is measured as the mean Euclidean distance from each feature vector to its class centroid, while interclass separability is quantified as the mean pairwise distance between all class centroids. Table IV and Fig. 10(b) and (c) show that SRNet achieves the best intraclass compactness of 0.1645 at T1 and maintains a similarly strong value of 0.1653 at T2, with an absolute temporal variation $|\Delta|$ of only 0.0008, which is substantially smaller than that of the second-best method BTSCD (|0.0398|). This near-zero drift indicates that SRNet's semantic feature representations remain highly consistent even under variations in illumination and object appearance—direct evidence that semantic drift has been effectively mitigated.

E. Ablation Comparison

As reported in Table V, we conducted ablation experiments across four datasets. SRNet achieves optimal performance on all metrics, while the removal of either MA or FGSM

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TABLE V
ABLATION STUDY ON SECOND, HI-UCD-MINI, SJH-SCD, AND HRSCD DATASETS

Model	SECOND			Hi-UCD-mini			SJH-SCD			HRSCD		
	SeK (%)	Fscd (%)	mIoU (%)	SeK (%)	Fscd (%)	mIoU (%)	SeK (%)	Fscd (%)	mIoU (%)	SeK (%)	Fscd (%)	mIoU (%)
SRNet	25.43	65.08	74.41	18.79	56.14	71.50	36.39	72.36	78.81	15.19	64.80	70.59
w/o MA	23.26	62.97	73.58	12.94	47.83	69.52	32.99	69.68	77.41	12.38	61.74	69.19
w/o FGSM	23.86	63.66	73.49	16.21	52.85	70.37	34.68	71.16	78.01	14.61	64.08	70.43
base	23.00	63.04	73.28	13.91	50.02	68.94	30.18	68.34	75.57	11.64	61.32	69.13

Note: SRNet denotes the full proposed model. *w/o MA* removes the Multi-Scale Fine-Grained Adapter, while *w/o FGSM* removes the Fine-Grained Spatiotemporal Module. *base* refers to the fully trained Swin backbone without any adapters.

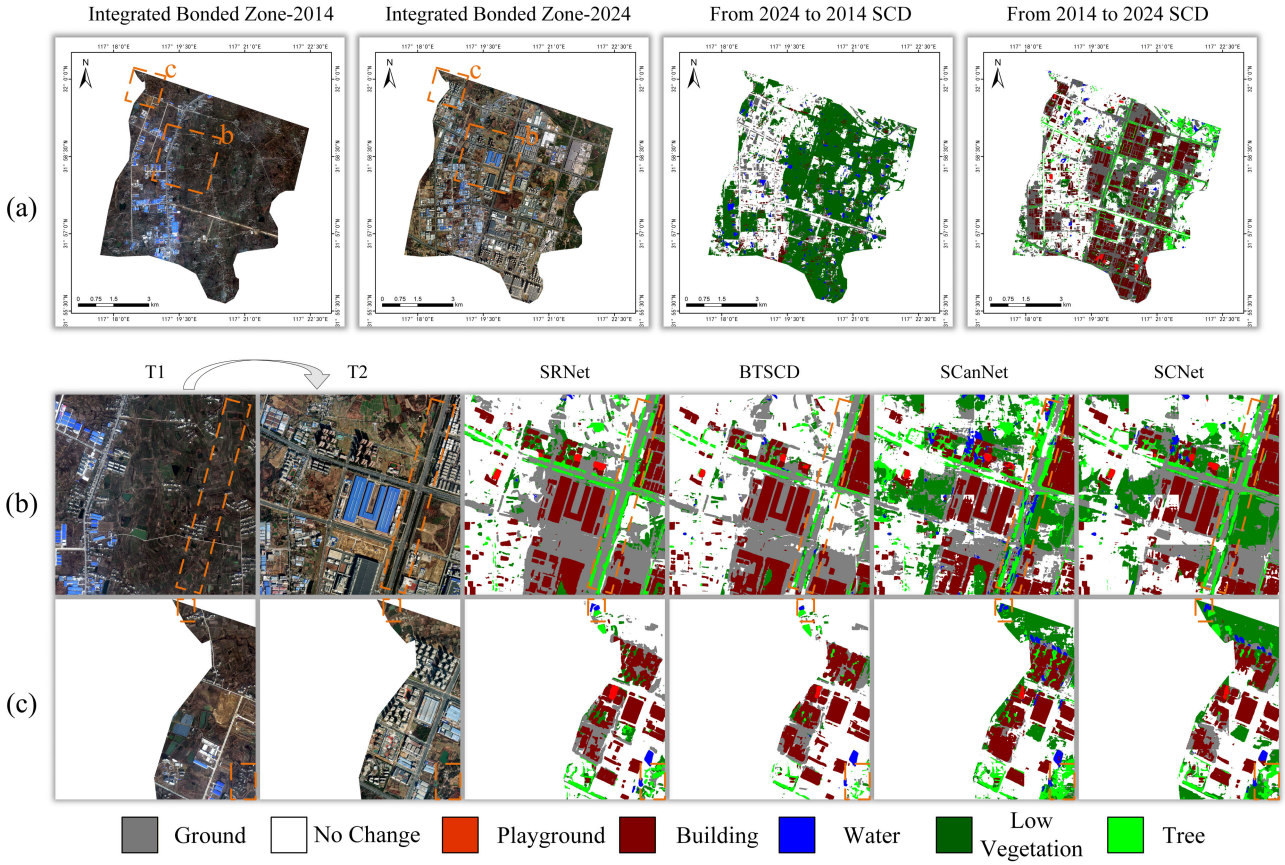


Fig. 12. (a) SCD results of Hefei comprehensive integrated bonded zone from 2014 to 2024 obtained by the SRNet model trained on the SECOND dataset. (b) and (c) Comparison of SRNet with other models on randomly selected regions.

leads to consistent degradation, confirming the individual effectiveness and complementary nature of the two modules.

To further illustrate their operational mechanisms, we present representative visualizations on the SECOND dataset in Fig. 11. The selected scenarios are characterized by high intraclass variance and low interclass variance, conditions notoriously prone to inducing false CDs. In these ambiguous scenarios, even the fully trained base model struggles to separate true land-cover transitions from spurious appearance shifts. As shown in the top-left portion of Fig. 11(c), the base model erroneously interprets a shift in roof color as a change from one building to another land-cover category. This failure reflects the model's inability to suppress spurious signals arising from intraclass variation. With the integration of FGSM, the network gains the capacity to explicitly couple cross-temporal cues and constrain semantic drift, thereby sharpening the boundary between true changes and false alarms. Consequently, SRNet correctly treats the roof color variation as unchanged. The MA module further strengthens semantic anchoring: its multiscale convolutions tighten features of the same class while pulling different classes apart, reducing semantic drift without sacrificing discriminability. We also compare the training dynamics of SRNet against a fully fine-tuned Swin-Base counterpart under identical settings. Despite updating only the MA modules and task heads, SRNet converges faster and reaches a higher final F_{scd}. The frozen backbone supplies stable semantic

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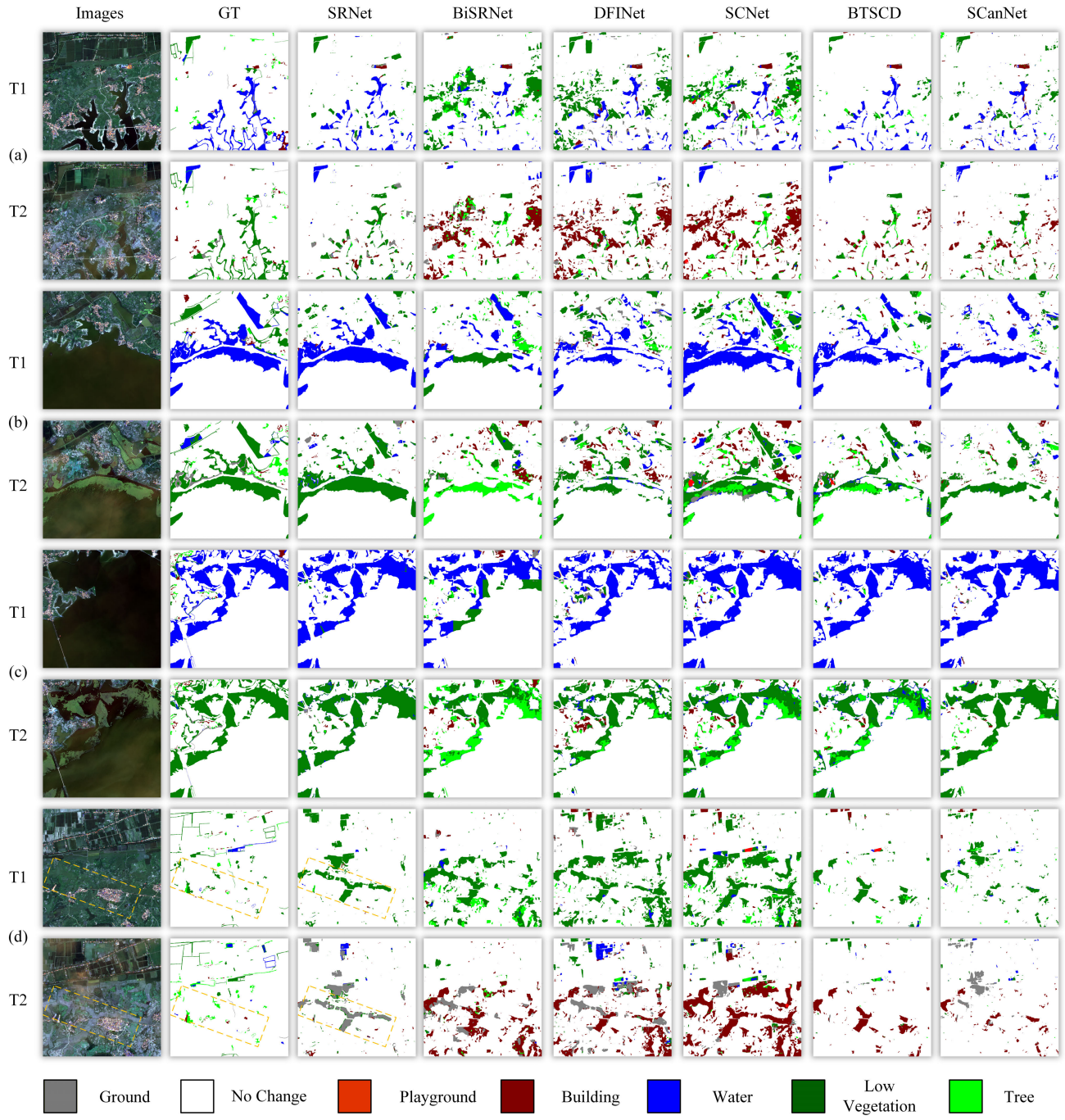


Fig. 13. (a)–(d) Cross-domain generalization results of SRNet trained on SECOND and tested on the SJH-SCD-7 dataset, respectively. SJH-SCD-7 is derived from SJH-SCD using the following class mapping: road/mudflat, ground → ground; aquatic vegetation/grassland, farmland → low vegetation; and forestland → tree.

priors, so the model does not waste capacity relearning general visual patterns from limited SCD data. Meanwhile, the lightweight MA modules adapt to domain-specific shifts without disturbing the backbone, keeping optimization stable. Taken together, MA-based PEFT saves trainable parameters, uses scarce labeled data more efficiently, and delivers stronger performance.

F. Generalization Analysis

1) Urban to Urban: As shown in Fig. 12, we further evaluate SRNet on 1-m high-resolution imagery of the Hefei Integrated Bonded Zone from 2014 to 2024. This area has experienced rapid urban development over the past decade, offering a representative testbed for assessing real-world generalization. The imagery was first partitioned into 2048 × 2048 patches. SRNet, initialized with weights trained on SECOND, was then applied to the 2014–2024 image pairs. The resulting patchwise predictions were mosaicked to produce the semantic change maps in Fig. 12(a). As shown, SRNet successfully delineates the decade-long land-cover transitions. Notable changes include the conversion from low vegetation to industrial zones, residential blocks,

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TABLE VI
CROSS-DOMAIN GENERALIZATION RESULTS ON THE SJH-SCD-7 DATASET. ALL MODELS ARE TRAINED ON THE SECOND DATASET AND DIRECTLY EVALUATED ON SJH-SCD-7 WITHOUT FINE-TUNING

Method	F_{scd} (%)	mIoU (%)	SeK (%)
BTSCD	49.41	66.56	8.38
SCanNet	49.20	62.48	5.37
SCNet	42.04	58.01	4.64
BiSRNet	25.54	53.29	-4.45
DFINet	36.08	53.90	-1.82
SRNet (Ours)	64.01	70.63	17.16

Note: Red indicates the best result, and blue indicates the second-best result.

and street trees, clearly reflecting the region's accelerated urbanization.

To further compare SRNet with mainstream SCD methods, we selected several competitive models trained on the SECOND dataset for qualitative analysis. Two representative subregions, denoted as (b) and (c) in Fig. 12, were randomly chosen for comparison. As highlighted by the yellow boxes, competing models exhibit considerable FPs and omissions in detecting elongated continuous tree changes as well as transitions among trees, water bodies, and low vegetation. This indicates the limitations of existing methods when transferred to regions with different data distributions. On the contrary, SRNet accurately identifies these changes and demonstrates superior performance in both local detail detection and global land-cover recognition. These results further validate that the proposed fine-tuning framework not only excels on a single dataset but also maintains strong generalization capability when transferred to other regions.

2) Urban to Wetland: To further challenge the generalization capability of SRNet, we conduct a cross-domain evaluation from urban to wetland scenes. In particular, we first consolidate the land-cover categories of the SJH-SCD dataset to produce SJH-SCD-7, ensuring semantic alignment with the SECOND dataset. The SRNet model trained on SECOND is then directly applied to SJH-SCD-7 without any fine-tuning. As shown in Fig. 13 and Table VI, SRNet performs favorably on several change types, including water bodies and low vegetation, producing coherent change boundaries without extensive false alarms. Quantitative results confirm this robustness, with SRNet achieving an mIoU of 70.63% and an Fscd of 64.01%. BiSRNet and DFINet obtain negative SeK scores; this negative SeK means the predicted change boundaries agree with the ground truth worse than random chance, indicating that the observed agreement between the predicted and reference change boundaries is lower than the agreement expected under randomness. Both methods are fully fine-tuned CNN-based architectures, which appear to overfit the urban spatial patterns in SECOND. As a result, their learned boundary representations fail to generalize to wetland scenes characterized by substantially different textures and spectral distributions, leading to severe degradation under cross-domain transfer.

Nevertheless, although SRNet achieves relatively favorable results, it still encounters difficulties in several challenging cases. For example, in Fig. 13(d), seasonal variations in paddy fields are misclassified as bare-land transitions. SJH-SCD-7 spans multiple seasons, involving variations in solar illumination and vegetation phenology. While the frozen Swin pretrained features provide partial invariance to seasonal changes, extreme seasonal or hydrological changes, such as snow cover and flooding, remain challenging for current SCD methods. Such errors are mainly attributable to the domain bias of the SECOND dataset, which primarily captures urban construction dynamics and provides limited representation of agricultural phenology. Addressing such cross-domain discrepancies remains a meaningful direction for future work.

V. CONCLUSION

This article presents SRNet, a PEFT framework for SCD. The main goal is to adapt large-scale pretrained visual representations to bitemporal remote sensing images while preserving their stable semantic priors. To this end, we introduce a semantic anchoring and fine-grained refinement strategy. The frozen pretrained backbone provides stable category-level semantics, while the proposed MA performs controlled task-specific adaptation around these semantic anchors. In addition, the FGSM further enhances subtle cross-temporal differences and multiscale change patterns, allowing the model to better distinguish true semantic changes from irrelevant appearance variations. Extensive experiments on four SCD datasets demonstrate the effectiveness of the proposed method across different scene types, including urban, rural, and wetland environments. SRNet achieves competitive or superior performance compared with existing methods while using only a small number of trainable parameters. Under the tested cross-domain setting, SRNet maintains competitive performance, indicating that semantic anchoring can provide a certain degree of robustness to distribution shifts. Comprehensive evaluation under broader cross-domain scenarios remains an important direction for future work. Moreover, the additional boundary quality analysis and feature-space analysis provide direct evidence that SRNet can alleviate semantic drift by improving boundary consistency and preserving more discriminative semantic representations.

Although the Swin-based MA adapter fine-tuning approach achieves significant progress in SCD, its backbone is pretrained only on the ImageNet-22k dataset, leaving the potential for larger-scale vision foundation models largely unexplored. Future work will further investigate the applicability of the MA adapter to other foundation models, aiming to leverage the powerful representations of large-scale pretrained models and promote continued advances in SCD.

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